4 Continuity

In the previous chapters, we developed a rigorous theory of convergent sequences and studied many of their applications. It is therefore natural to ask whether convergence is preserved under a function. More precisely, if (x_n) is a sequence converging to a point a, under what conditions does the sequence $\big(f(x_n)\big)$ converge to $f(a)$? Functions satisfying this property are called *continuous functions*. This sequential viewpoint provides a natural and elegant introduction to continuity and builds directly on the ideas developed earlier.

Intuitively, continuity means that small changes in the input produce only small changes in the output; in other words, there are no sudden jumps or breaks. Many quantities encountered in nature, such as temperature, speed, or pressure, vary continuously over time. Geometrically, a continuous function can often be visualized as one whose graph can be drawn from left to right without lifting the pen from the paper.

4.1 Continuous functions

Definition 27. *Let $J \subseteq \mathbb{R}$, let $f : J \to \mathbb{R}$ be a function, and let $a \in J$. We say that f is* continuous at a *if for every sequence (x_n) in J with $x_n \to a$, we have $f(x_n) \to f(a)$. We say that f is* continuous on J *if it is continuous at every point of J.*

Example 62. *Show that every constant function is continuous.*
Solution. *Let $J \subseteq \mathbb{R}$ and let $f : J \to \mathbb{R}$ be the constant function defined by $f(x) = c$ for all $x \in J$.*
Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \to a$. Since $f(x_n) = c$ for every n, the sequence $\big(f(x_n)\big)$ is constant. Hence,

$$f(x_n) \to c = f(a).$$

Therefore, f is continuous at a. As a was arbitrary, we conclude that f is continuous on J.

Example 63. *Show that the identity function $f : J \to \mathbb{R}$, defined by $f(x) = x$ for all $x \in J$, is continuous.*
Solution. *Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \to a$. Since $f(x_n) = x_n$ for every n, we have*

$$f(x_n) = x_n \longrightarrow a = f(a).$$

Hence, f is continuous at a. Since a was arbitrary, f is continuous on J.
More generally, for every $n \in \mathbb{N}$, the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^n$ is continuous.

Example 64. *Let $f : \mathbb{R} \to \mathbb{R}$ be defined by*

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then f is not continuous at 0.
Solution. *Consider the sequence $x_n = \dfrac{1}{n}$. Then $x_n \in \mathbb{Q}$ and $x_n \to 0$. Hence*

$$f(x_n) = 1 \longrightarrow 1 = f(0).$$

Now let

$$y_n = \frac{\sqrt{2}}{n}.$$

Since $\sqrt{2}$ is irrational, y_n is irrational for every n. Also, $y_n \to 0$. Therefore,

$$f(y_n) = 0 \longrightarrow 0 \neq 1 = f(0).$$

Thus, by the sequential criterion for continuity, f is not continuous at 0.
Using the density of the rational and irrational numbers in $\mathbb{R}$, together with the Sandwich Theorem, one can similarly show that f is not continuous at any point of $\mathbb{R}$.

Example 65. *Consider the function*

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

We claim that f is continuous at every $a \neq 0$ but is not continuous at 0.
Indeed, let $a \neq 0$ and let (x_n) be a sequence converging to a. Since a is either positive or negative, there exists $N \in \mathbb{N}$ such that x_n has the same sign as a for all $n \geq N$. Consequently, $f(x_n) = f(a)$ for all sufficiently large n, and hence

$$f(x_n) \longrightarrow f(a).$$

Thus f is continuous at every $a \neq 0$.
Now let $a = 0$. Consider the sequences

$$x_n = \frac{1}{n}, \qquad y_n = -\frac{1}{n}.$$

Both converge to 0, but

$$f(x_n) = 1 \longrightarrow 1, \qquad f(y_n) = 0 \longrightarrow 0.$$

Since the limits are different, f is not continuous at 0.

Example 66. *Let $f : \mathbb{R} \to \mathbb{R}$ be defined by*

$$f(x) = \begin{cases} \alpha, & x < 0, \\ ax^2 - bx + c, & x \geq 0. \end{cases}$$

Determine the value of α for which f is continuous at 0.
Solution. *Suppose that f is continuous at 0. Consider the sequence*

$$x_n = -\frac{1}{n}.$$

Then $x_n \to 0$ and

$$f(x_n) = \alpha \longrightarrow f(0) = c.$$

Hence,

$$\alpha = c.$$

Conversely, assume that $\alpha = c$. Let (x_n) be any sequence with $x_n \to 0$. If $x_n < 0$, then

$$f(x_n) = c = f(0).$$

If $x_n \geq 0$, then

$$f(x_n) - f(0) = ax_n^2 - bx_n \longrightarrow 0,$$

by the algebra of convergent sequences. Therefore,

$$f(x_n) \longrightarrow f(0).$$

Hence, f is continuous at 0. Thus, f is continuous at 0 if and only if

$$\boxed{\alpha = c.}$$

Theorem 28 (Algebra of Continuous Functions)**.** *Let $J \subseteq \mathbb{R}$, let $f, g : J \to \mathbb{R}$ be continuous at $a \in J$, and let $\alpha \in \mathbb{R}$. Then:*

1. *$f + g$ is continuous at a.*
2. *αf is continuous at a.*
3. *fg is continuous at a.*
4. *If $g(a) \neq 0$, then there exists $\delta > 0$ such that $g(x) \neq 0$ for all $x \in (a-\delta, a+\delta) \cap J$. Then $\dfrac{1}{g}$ is continuous at a.*

5. $|f|$ *is continuous at* a.

6. *The functions* $\max\{f, g\}$ *and* $\min\{f, g\}$ *are continuous at* a.

7. *If* $K \subseteq \mathbb{R}$, $h : K \to \mathbb{R}$ *is continuous at* $f(a)$, *and* $f(J) \subseteq K$, *then the composition* $h \circ f$ *is continuous at* a.